HOMEWORK 1
BASICS, CATEGORIES, GROUPS AND EXAMPLES

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Disclaimer: Generative AI was used to help me find a decent way to typeset a permutation and to determine whether I should say an element is an equivalence class "over" or "under" a relation.

1. In general, for $X = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathrm{Mat}_2(\mathbb{R})$, we have

$$XM = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} 2p + r & 2q + s \\ 2r & 2s \end{pmatrix}$$

and

$$MX = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2p & p + 2q \\ 2r & r + 2s \end{pmatrix},$$

so the matrices X which satisfy $MX = XM$ (i.e. $X \in \mathcal{B}$) are exactly those which solve the following system of equations:

$$\begin{aligned} 2p + r &= 2p \\ 2q + s &= p + 2q \\ 2r &= 2r \\ 2s &= r + 2s. \end{aligned}$$

The third equation tells us nothing, but the first and fourth equations imply that $r = 0$, while the second says that $s = p$. This leaves us with $X = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix}$. This is necessary since we derived it from the general form of a 2×2 matrix X, but it is also sufficient since plugging the values back into the system of equations yields

$$\begin{aligned} 2p + 0 &= 2p \\ 2q + p &= p + 2q \\ 0 &= 0 \\ 2p &= 0 + 2p. \end{aligned}$$

$\square$

Challenge: INCOMPLETE

2.

(a) Consider $a = 4, b = 6$. Then $f\left(\frac{a}{b}\right) = f\left(\frac{4}{6}\right) = 4$, but using the fact that $\frac{a}{b} = \frac{4}{6} = \frac{2}{3}$, we have $f\left(\frac{a}{b}\right) = f\left(\frac{2}{3}\right) = 2$. Thus f is not well-defined. $\square$

(b) Since $\frac{a^2}{b^2} = \left(\frac{a}{b}\right)^2$, this is equivalent to saying that $f(x) = x^2$ for any $x \in \mathbb{Q}$. This is unambiguous, and in particular it is not dependent on the representation of x as a ratio of two integers like part (a), so it is well-defined. $\square$

3. If $a, b, c \in X$, then $f(a) = f(a)$, so $a \sim_f a$. Also, if $a \sim_f b$, then $f(a) = f(b)$ and $f(b) = f(a)$, so $b \sim_f a$. Finally, if $a \sim_f b$ and $b \sim_f c$, then $f(a) = f(b) = f(c)$, so $a \sim_f c$. In short, $\sim_f$ "inherits" its reflexivity, symmetry, and transitivity from "=".

If $f^{-1}(\{y\})$ is nonempty for some $y \in Y$, then for $a, b \in f^{-1}(\{y\})$, we have $f(a) = f(b) = y$, so $a \sim_f b$. Conversely, given some $a, b \in X$ where $a \sim_f b$, take $y = f(a) = f(b)$. Then $a, b \in f^{-1}(\{y\})$, so the equivalence classes under $\sim_f$ are exactly the nonempty fibers of y.

Finally, given an equivalence relation $\sim$ on a set X, define $f : X \to X/\sim$ by $f(x) = [x]_\sim$ for any $x \in X$, where $[x]_\sim$ is the equivalence class of x under $\sim$. Then f is surjective since $[y]_\sim = f(y)$ for any $[y]_\sim \in X/\sim$.

If $a, b \in X$ such that $a \sim b$, then $f(a) = [a]_\sim = [b]_\sim = f(b)$, so $a \sim_f b$. On the other hand, if $a \sim_f b$, then $f(a) = f(b) = [b]_\sim = [a]_\sim$, so $a \sim b$ since they have the same equivalence class under $\sim$.

4. First, note that the prime factorization are $69 = 3 * 23$ and $372 = 2 * 2 * 3 * 31$. Thus the greatest common divisor of 69 and 372 is 3. Take $a =$ and $b =$. Then

$$69a + 372b = 69 * +372* =$$

UNFINISHED

5.

6. Given $x \in \mathbb{Z}$, if x is even, then $x = 2n$ for some integer n, and $x^2 = 4n^2$, so $x^2 \equiv 0 \bmod 4$. If x is odd, then $x = m + 1$ for some even m. Note that $2m \equiv 0 \bmod 4$ since m is even, so

$$\begin{aligned} x^2 &\equiv m^2 + 2m + 1 \pmod 4 \\ &\equiv 0 + 0 + 1 \pmod 4 \\ &\equiv 1 \pmod 4. \end{aligned}$$

In general, then, x^2 is congruent to either 0 or 1 mod 4 for any integer x.

Now consider $a^2 + b^2 = 3c^2 \bmod 4$. For nonzero $a, b, c \in \mathbb{Z}$, since c^2 is congruent to either 0 or 1 mod 4, $3c^2$ is congruent to either 0 or 3 mod 4. The left-hand side of the equation mod 4 is either $0 + 0$, $0 + 1$, $1 + 0$, or $1 + 1$, so it cannot be 3 mod 4, and thus it must be 0. The only way for this to happen is to have $a^2 \equiv b^2 \equiv 0 \bmod 4$, so we can factor out a 4 to get

$$4a_0^2 + 4b_0^2 = 3(4c_0^2)$$

where a_0, b_0, and c_0 are integers. Dividing by 4 yields $a_0^2 + b_0^2 = 3c_0^2$, revealing another solution to the equation which, as we have shown, must be congruent to 0 mod 4 on both sides. Since the prime factorization of an integer is of finite length, however, we can repeat this process of dividing by 4 until we reach an equation where the sides are not congruent to 0 mod 4, which we have shown to be impossible.

$\square$

7.

8.

9. Indeed, $x = x^{-1}$ for all $x \in G$ since $x * x = 1$, so given $g, h \in G$,

$$\begin{aligned} gh &= (gh)^{-1} \\ &= h^{-1}g^{-1} \\ &= hg. \end{aligned}$$

$\square$

10. (a)

$$\begin{aligned} v^3 = 1 &\implies v^3 * v^{-1} = 1 * v^{-1} \\ &\implies v^2 * (v * v^{-1}) = v^{-1} \\ &\implies v^2 * 1 = v^{-1} \\ &\implies v^2 = v^{-1} \end{aligned}$$

(b) First, observe that

$$\begin{aligned} v^2u^3v &= (v^2u^2)uv \\ &= (uv)uv \\ &= uv(uv) \\ &= uv(v^2u^2) \\ &= uv^3u^2 \\ &= u^3. \end{aligned}$$

Multiplying by v on the left, we get

$$\begin{aligned} &v^3u^3v = vu^3 \\ \implies &u^3v = vu^3. \end{aligned}$$

(c) First, we establish that

$$\begin{aligned} u^9 &= (u^8)u \\ &= (u^4u^4)u \\ &= u, \end{aligned}$$

from which we derive

$$\begin{aligned} vu &= vu^9 \\ &= (vu^3)u^3u^3 \\ &= u^3(vu^3)u^3 \\ &= u^3u^3(vu^3) \\ &= u^3u^3u^3v \\ &= u^9v \\ &= uv. \end{aligned}$$

(d) Since u and v commute, we can transform the "commuting" relation like so:

$$\begin{aligned} uv &= v^2u^2 \\ &= u^2v^2 \\ &= u(uv)v \\ &= (uv)(uv) \\ &= (uv)^2. \end{aligned}$$

Multiplying by $(uv)^{-1}$ on both sides yields $1 = uv$.

(e) Indeed,

$$\begin{aligned} u &= u(1)^3 \\ &= u(uv)^3 \\ &= u^4v^3 \\ &= 1 \cdot 1 \\ &= 1. \end{aligned}$$

Then since $uv = 1$, substituting 1 for u yields $v = 1$.

Rewriting the presentation in light of this, we have

$$\begin{aligned} G &= \left\langle u, v \mid u^4 = v^3 = 1, uv = v^2u^2 \right\rangle \\ &= \left\langle 1, 1 \mid 1^4 = 1^3 = 1, 1 \cdot 1 = 1^2 1^2 \right\rangle \\ &= \langle 1 \rangle \end{aligned}$$

where the third equality comes from removing duplicate generators and trivial relations. Thus, G is the trivial group.

$\square$

11.

12.